A Note on Coupled Oscillators (sample document)

Two pendulums hung from a common beam slowly synchronize. This invented note sketches the phase-response picture: each oscillator nudges the other's phase by an amount that depends on where in its cycle the nudge arrives.

Section 1. The phase model: dtheta_i/dt = omega_i + K * sin(theta_j - theta_i).
Section 2. Locking occurs when |omega_1 - omega_2| < 2K.